

Name: Solutions (please print)
Signature: Solutions

ECE 2200 – Quiz 0

January 15, 2025

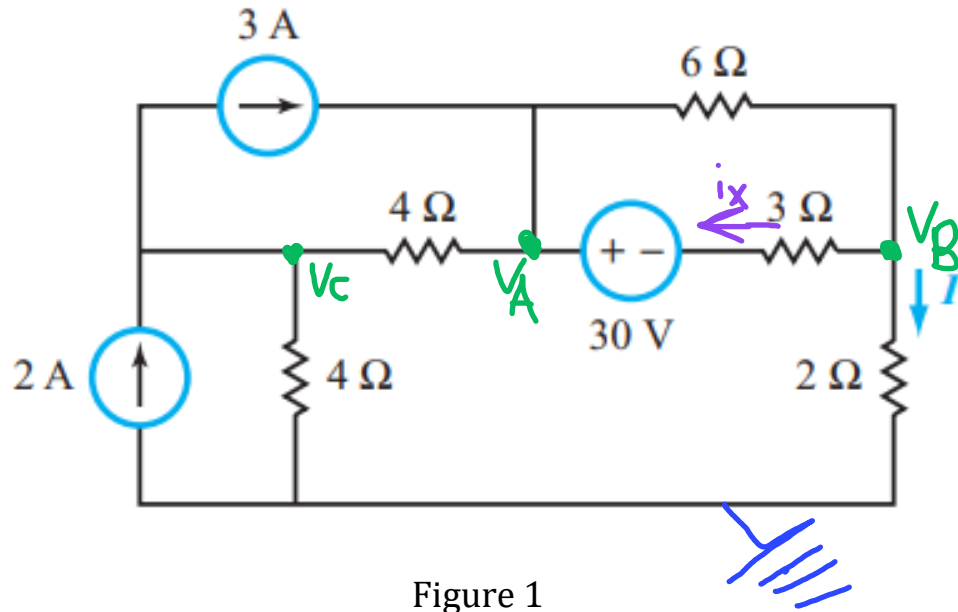
1. Show all work necessary to complete the problem. Use additional sheets of paper as needed.
3. Show all units in solutions and figures. Units in the quiz will be included between square brackets.
5. You will have 60 minutes to work on the quiz. For online students: this time includes the time to print, scan, and upload your work. Points will be deducted if it is not uploaded by the deadline time.

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Problem 1

In Figure 1,

- Find current I using the **node voltage method**.
- What is the power delivered by the 30 [V] source?



Problem 1

a) Node V_A : $\frac{V_A - V_C}{4} + \frac{V_A - V_B - 30}{3} - 3 + \frac{V_A - V_B}{6} = 0$

Node V_B : $\frac{V_B - V_A + 30}{3} + \frac{V_B}{2} + \frac{V_B - V_A}{6} = 0$

Node V_C : $\frac{V_C - V_A}{4} + \frac{V_C}{4} - 2 + 3 = 0$

Solve V_A, V_B, V_C $V_A = 20[V], V_B = 0[V], V_C = 8[V]$

Ohm's Law: $I = \frac{V_B}{2}$ $I = 0[A]$

b) $P_{del. by 30[V] source} = i_x \cdot 30$

$i_x = \frac{V_B - V_A + 30}{3} = 3.3\bar{3}[A]$

$P_{del. by 30[V] source} = 3.3\bar{3} \cdot 30 =$ $100[W]$

Problem 2

Repeat Problem 1 part a (i.e., find current I) but apply the **source transformation method** instead. Figure 1 is shown below again for convenience.

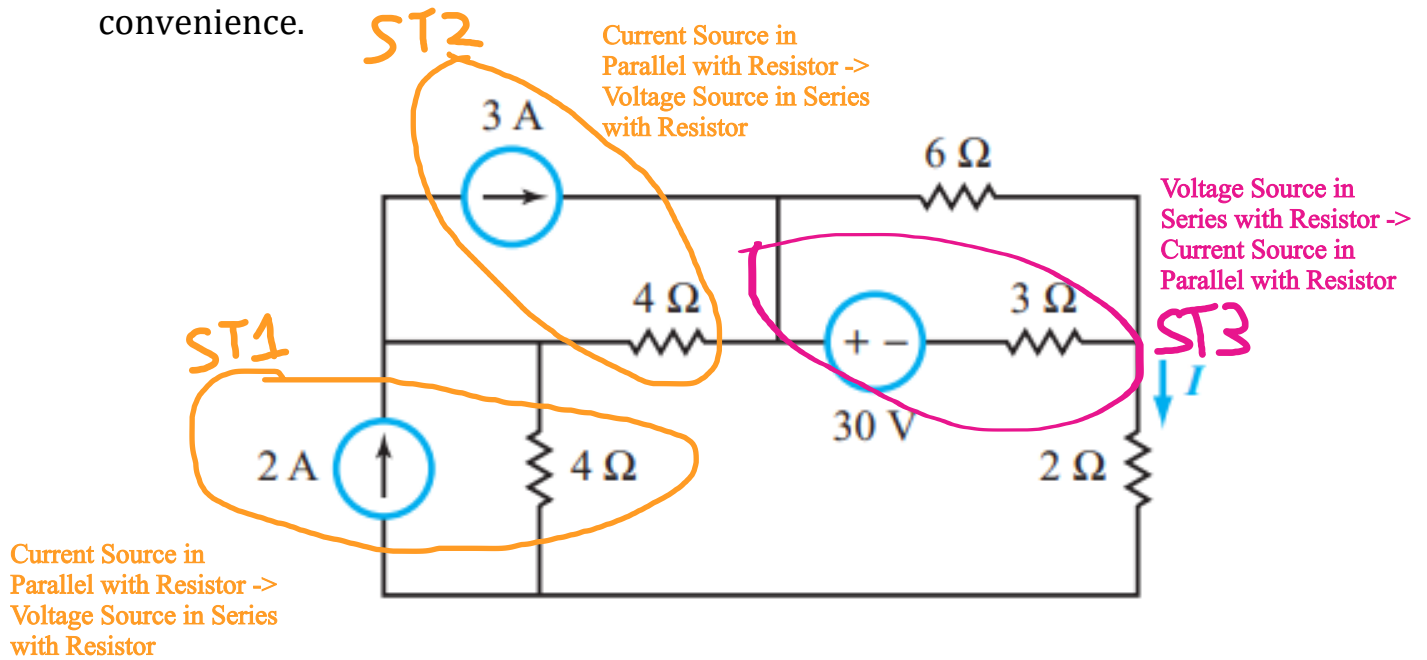
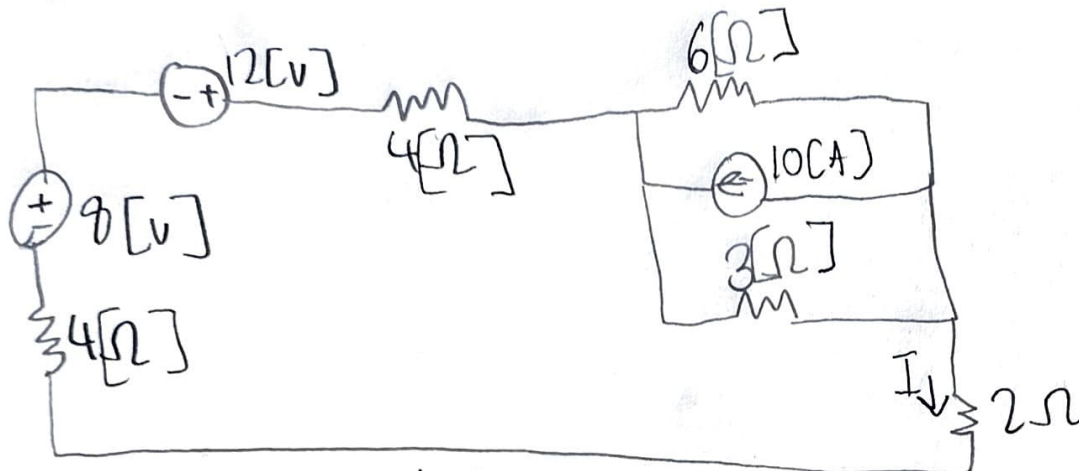
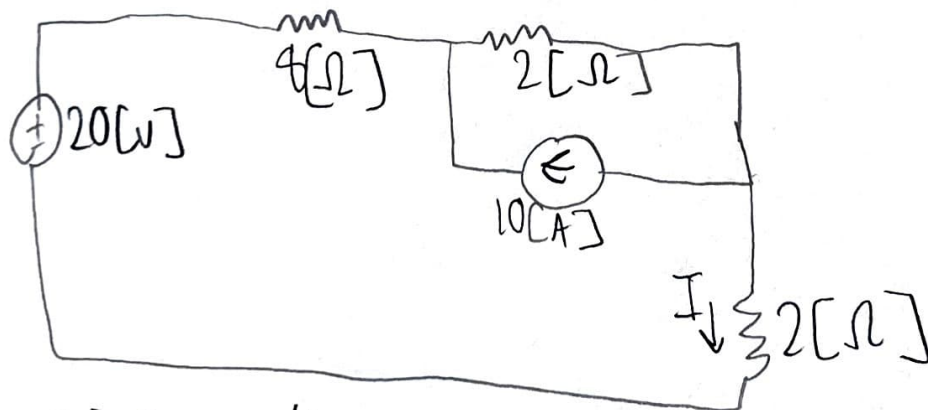


Figure 1

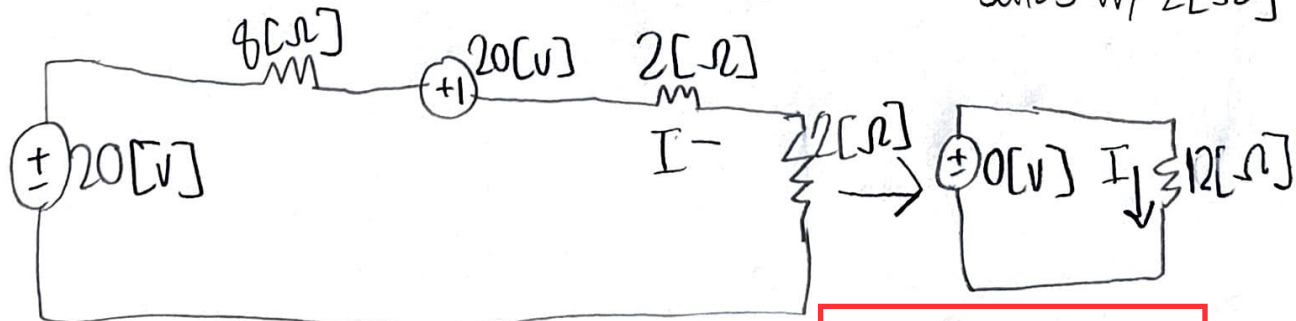
2) ST1: $V_{\text{source}} = 2 \cdot 4 = 8 \text{ [V]}$
 ST2: $V_{\text{source}} = 3 \cdot 4 = 12 \text{ [V]}$
 ST3: $C_{\text{source}} = \frac{30}{3} = 10 \text{ [A]}$



Resistors in series $(4\Omega + 4\Omega) = 8\Omega$
 $\downarrow$ V_{source} in series $(8\text{V} + 12\text{V}) = 20\text{V}$
 Resistors in parallel $(6\Omega // 3\Omega) = 2\Omega$



ST: $10\text{ [A]} // 2\Omega \Rightarrow 20\text{ [V]}$ V_{source} in series w/ 2Ω



$$I = \frac{0}{12} = 0 \text{ [A]}$$

Problem 3

For Figure 2, obtain an expression for the voltage gain $G=v_o/v_s$.

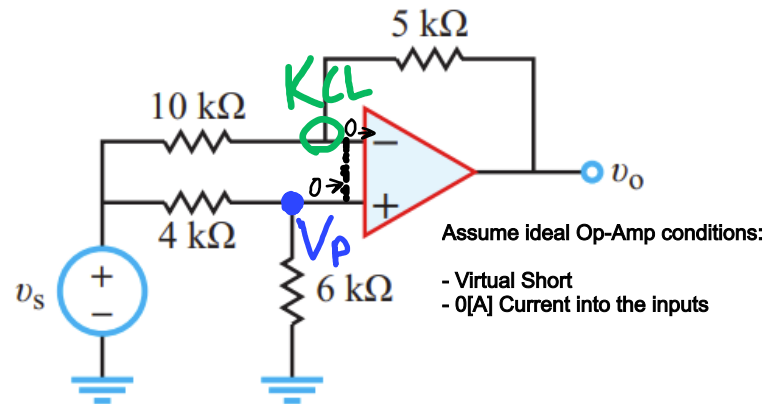


Figure 2

3) $4 [k\Omega]$ & $6 [k\Omega]$ are technically in series,
Since $O[A]$ current goes into the + input.
Therefore, we may use Voltage Divider to get V_p
in terms of V_s .

$$V_p = \left(\frac{6000}{6000+4000} \right) \cdot V_s = \frac{3}{5} V_s, \quad V_p = \frac{3}{5} V_s$$

Green KCL: $\frac{\frac{3}{5} V_s - V_s}{10000} + \frac{\frac{3}{5} V_s - V_o}{5000} = 0$

Isolate $\frac{V_o}{V_s}$: $\downarrow \times 10000$

$$\frac{3}{5} V_s - V_s + 2 \left(\frac{3}{5} V_s - V_o \right) = 0$$

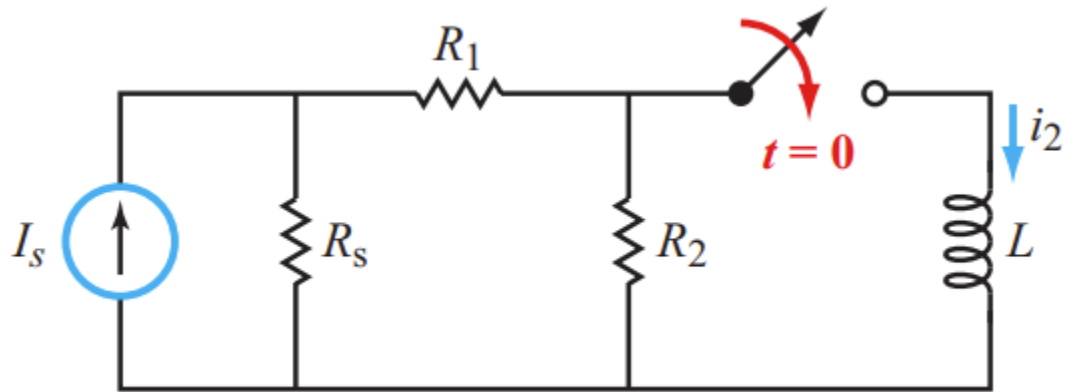
$$-\frac{2}{5} V_s + \frac{6}{5} V_s - 2 V_o = 0$$

$$\frac{4}{5} V_s = 2 V_o$$

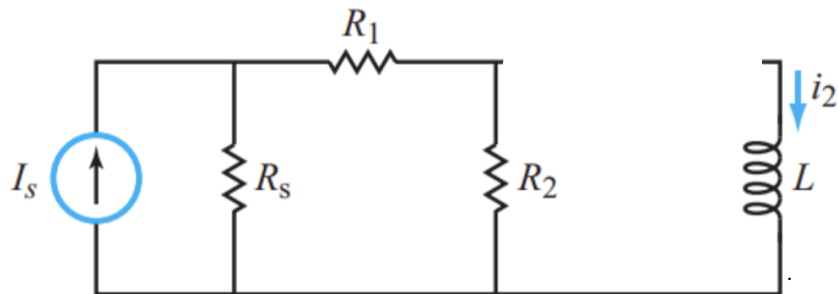
$$\boxed{\frac{V_o}{V_s} = \frac{4}{10} = 0.4}$$

Problem 4

Write an expression for $i_2(t)$ for $t \geq 0$.

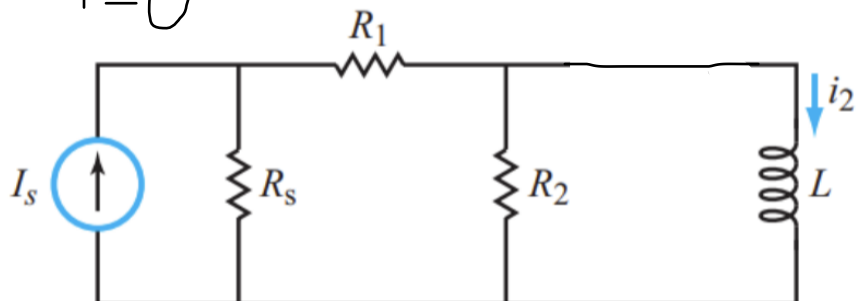


$t < 0$



$$i_2(0) = 0[A]$$

$t \geq 0$



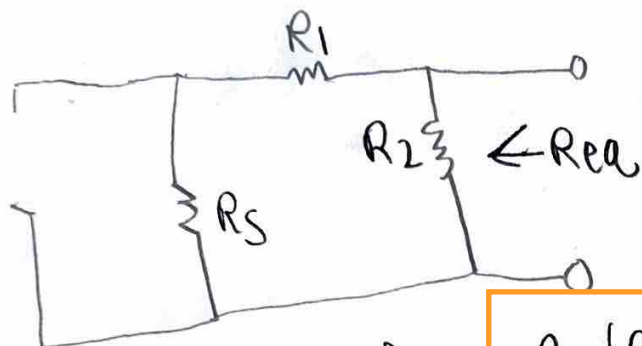
Step Response

4)
$$i_2(t) = i_2(\infty) + [i_2(t_0) - i_2(\infty)] e^{-\frac{(t-t_0)}{\tau}}$$

For inductors, $\tau = L/R_{eq}$ $t_0 = 0$

$$i_2(0) = 0 \text{ [A]}$$

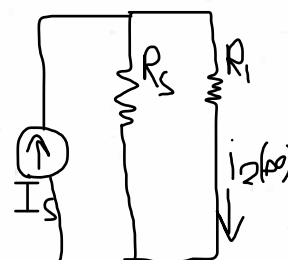
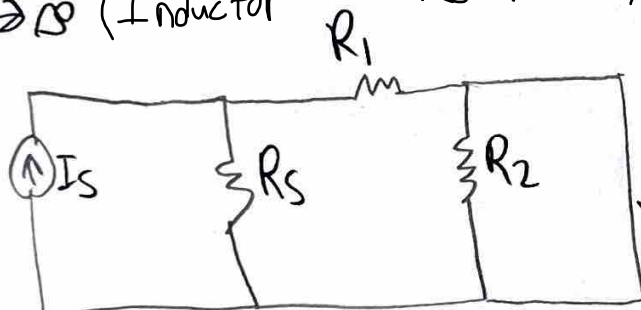
Find R_{eq} :



$$R_{eq} = R_2 \parallel (R_1 + R_S) = \frac{R_2 (R_1 + R_S)}{R_2 + R_1 + R_S}$$

Find $i_2(\infty)$:

$t \rightarrow \infty$ (Inductor acts like a short)



R_2 is shorted. R_1 and R_S are in parallel. We can use Current Divider Rule

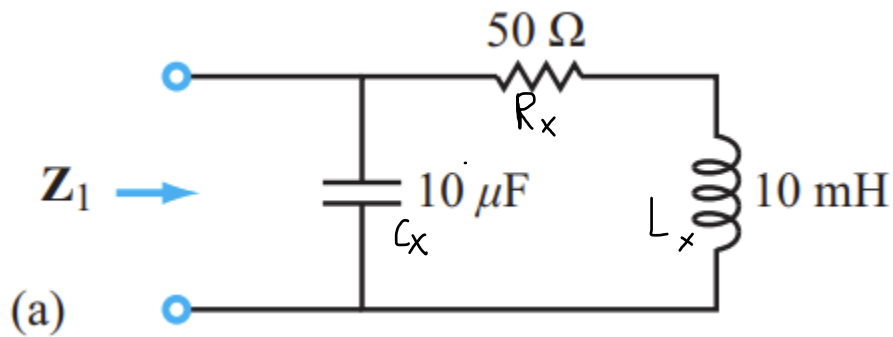
$$i_2(\infty) = I_S \left(\frac{R_S}{R_1 + R_S} \right)$$

For $t \geq 0$

$$i_2(t) = i_2(\infty) - i_2(\infty) e^{-\frac{t \cdot R_{eq}}{L}}$$

Problem 5

Find the equivalent impedance Z_1 at 1000 [Hz]



Convert to Phasor Domain:

Component	Value	Transform
Voltages	$v_X(t) = V_m \cos(\omega t + \phi_v)$	$V_{xm}(\omega) = V_m \angle \phi_v$
Currents	$i_X(t) = I_m \cos(\omega t + \phi_i)$	$I_{xm}(\omega) = I_m \angle \phi_i$
Resistors	R_X	$Z_{R_X} = R_X$
Inductors	L_X	$Z_{L_X} = j\omega L_X$
Capacitors	C_X	$Z_{C_X} = 1/j\omega C_X$

$$\cancel{1/j\omega C_X} = \frac{-j}{\omega C_X}$$

Problem 5

$$f = 1000 [\text{Hz}]$$

$$\omega = 2\pi f = 2000\pi [\text{rad/s}]$$

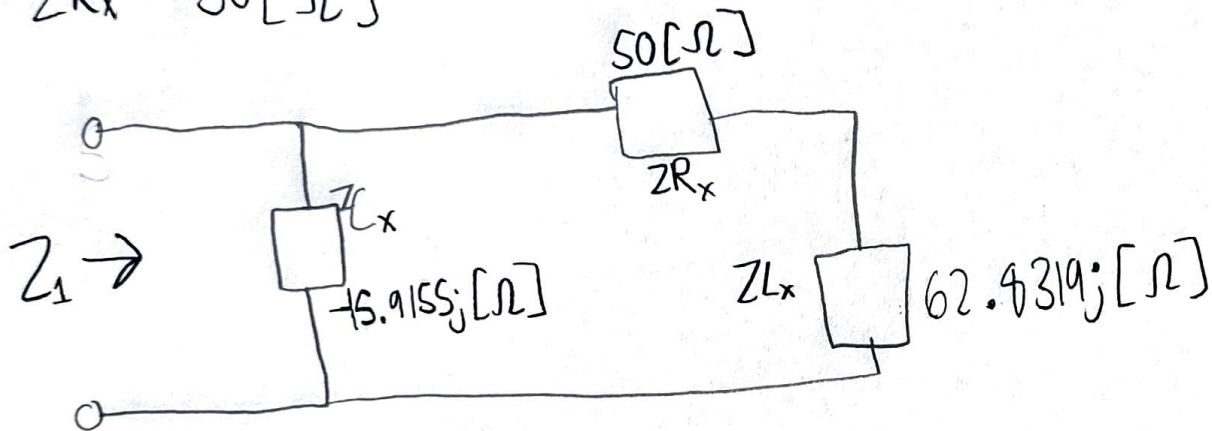
$$C_x = 10 \times 10^{-6} [\text{F}] \quad L_x = 10 \times 10^{-3} [\text{H}] \quad R_x = 50 [\Omega]$$

Convert to Phasor Domain:

$$Z_{C_x} = \frac{-j}{\omega C_x} = \frac{-j}{2000\pi \cdot 10 \cdot 10^{-6}} = -15.9155j [\Omega]$$

$$Z_{L_x} = j\omega L_x = 62.8319j [\Omega]$$

$$Z_{R_x} = 50 [\Omega]$$



Simplify impedances:

$$Z_1 = Z_{C_x} // (Z_{R_x} + Z_{L_x}) = \frac{(-15.9155j)(50 + 62.8319j)}{-15.9155j + 50 + 62.8319j}$$

$$Z_1 = 2.69406 - 19.4434j [\Omega]$$